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Inverse design of terahertz amplitude modulator using tandem deep neural networks

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ABSTRACT

The terahertz (THz) frequency range has emerged as a promising spectral window for broad applications, including next-generation wireless communication, high-resolution imaging, and ultrafast spectroscopy. Among the essential components in these systems, amplitude modulators with high quality (Q) factors can provide sharp, selective frequency responses, which are key requirements for scalable and high-performance THz systems. However, designing high-Q THz modulators remains challenging, as conventional full-wave simulations are time-consuming and inefficient. In this study, we propose a deep learning-based inverse design framework tailored for THz metasurfaces composed of split-ring resonators (SRRs). The framework is built on a tandem neural network architecture that couples a forward model with an inverse network to retrieve structural parameters from desired spectral responses. To enhance physical feasibility and predictive stability, we introduce an autoencoder-based spectral projection method. Our model accurately reconstructs SRR geometries across a wide range of spectral targets by learning the underlying physical relationships. Notably, we demonstrate the inverse design of Fano resonant geometries characterized by high-Q factors and sharp asymmetric resonances, which are essential features for achieving deep modulation. By extending the tandem deep learning approach to the THz domain and incorporating an autoencoder-based spectral projection, our framework provides a scalable and efficient pathway for the rapid prototyping of tunable, high-Q THz devices and lays the foundation for artificial intelligence-driven design of advanced THz photonic components.

The terahertz (THz) spectral region, spanning approximately 0.1–10 THz, bridges the gap between microwave and optical frequencies and offers a distinctive property that enables a broad range of scientific and technological applications, including high speed 6G communication, nondestructive imaging, and ultrafast spectroscopy.[1–5] To fully exploit the potential of THz technology, the development of actively tunable devices is essential. Among them, THz amplitude modulators that regulate the intensity of transmitted or reflected THz waves play a key role in adaptive imaging systems, reconfigurable communication networks, and programmable sensing platforms.[6–8] Among the various modulator architectures, split-ring resonator (SRR) structures have emerged as highly effective and widely adopted solutions.[9–13] They enable efficient THz modulation through diverse physical mechanisms, including inductive-capacitive (LC) coupling, Fano resonance, electromagnetically induced transparency (EIT), and bound states in the continuum (BIC).[14–20] Despite their advantages, conventional design methods rely on full-wave electromagnetic simulations, extensive parameter sweeps, and manual optimization, which are computationally expensive, time-consuming, and increasingly impractical as structural complexity grows.

To address these challenges, inverse design methods have been widely employed to determine optimal structural parameters in a variety of nanophotonic applications, including metasurfaces, photonic crystals, and silicon optical waveguides.[21–24] These approaches enable physics-aware optimal design of structures that meet specific spectral requirements, such as resonance wavelength, bandwidth, and modulation depth, through computer-assisted design workflows, thereby replacing manual parameter sweeps with data-driven optimization.[25–27] Despite the growing importance of the THz band, artificial intelligence (AI)-driven structural design, particularly inverse design, remains relatively underexplored compared to its application in other frequency domains.[28–31]

In this work, we present a tandem network-based inverse design framework tailored for THz amplitude modulators employing SRR architectures. This framework streamlines the device design process by offering a robust, time-efficient, and accurate means of retrieving structural parameters from desired spectral responses. By coupling a pretrained forward network with an inverse network, the tandem architecture reduces convergence instability and effectively handles the inherent non-uniqueness of the inverse mapping that often limits direct inverse models.[32–34] To further enhance the reliability of inverse predictions, we incorporate an autoencoder-based spectrum projection strategy that transforms arbitrary defined target spectrum into a physically plausible spectral profile prior to inverse retrieval. In particular, we demonstrate the inverse design of SRR structures that exhibit Fano resonance characteristics, featuring sharp asymmetric spectral profiles and strong field localization, to highlight the model's capability in designing high Q and physically interpretable modulators in a simple manner. We validate our approach across diverse target spectra, demonstrating that the inverse model predicts SRR structures that yield modulation spectra closely matching the intended spectral features. Additionally, the model's ability to capture multi-solutions inherent to the inverse problem confirms that it learns the underlying physical relationship between spectral features and SRR geometries, rather than merely memorizing the training data. By extending the tandem deep learning approach to the THz domain, our framework significantly reduces design time and facilitates rapid prototyping of custom THz metasurfaces, thereby advancing the practical deployment of THz photonic systems.

In the THz frequency regime, active modulation of electromagnetic waves is an essential function for developing high speed communication, sensing, and imaging systems. Among various metastructure-based modulators, SRR structures are among the most representative structures due to their highly tunable electromagnetic response, which can be precisely engineered by adjusting the geometric configuration. We define the SRR structure using eight geometric parameters: G_n, number of gaps; P, unit cell periodicity; S_w, structure width; L_w, linewidth; G_d, gap distance; G_{p_1st}, position of the first gap; G_{p_2nd}, position of the second gap; G_{p_r}, gap rotation [Fig. 1(a)]. The modulation characteristics across the THz band can be sufficiently tuned by these parameters, enabling precise control over the spectral response. A total of 13 000 samples were generated, each defined by eight geometrical parameters and paired with its corresponding transmittance spectrum sampled at 1001 frequency points across the 0.1–1.5 THz range. To enable both forward and inverse mappings between SRR geometries and their corresponding THz spectra, we trained the model using a paired dataset consisting of FDTD-simulated transmittance spectra and their corresponding structural parameters (see the supplementary material, Sec. 1).

The tandem neural network architecture employed for THz modulator design consists of two core components: a forward prediction network and an inverse design network [right panel in Fig. 1(b)]. In the forward network, the structural parameters are provided as input, and the corresponding spectral response is produced as output based on the trained dataset. In contrast, in the inverse network, a target spectrum is used as input, and the predicted structural parameters are returned as output. Both networks are implemented as multilayer fully connected (FC) architectures. The forward model is designed with progressively increasing feature dimensionality, while the inverse model adopts a mirrored structure with decreasing dimensionality. This bidirectional configuration enables accurate modeling of the nonlinear mapping between the physical parameters of the metasurface and their corresponding THz transmittance spectra.

Each data sample comprises a $p = [p_1, p_2, \ldots, p_n,]$ representing the geometry of the SRR. The corresponding spectral transmittance $S = [S(f_1),\ S(f_2), \ldots, S(f_m)]$, which is calculated across the THz f_m, denotes each spectral point. Each structural configuration is p represented as a 1×8 vector S, and the corresponding transmittance spectrum is represented as a 1×1001 vector S.

The structural parameters were standardized using Z-score normalization, which is a widely used technique in machine learning that transforms the data so that the mean of each feature becomes zero and the standard deviation becomes one. These preprocessing steps mitigate scale disparities among the structural parameters, promoting more balanced gradient updates during optimization.[35,36] In contrast, the transmittance spectra, which are inherently bounded between 0 and 1, were used without further normalization to preserve their physical meaning, which is critical for accurate inverse design. As a result, the model avoids biased learning toward high magnitude features and exhibits improved training stability and faster convergence across diverse design configurations. Paired structural parameters and their corresponding spectra are utilized as the supervised training data for both the forward and inverse networks,

$$\hat{S} = \mathrm{ForwardNet}(p), \tag{1}$$

where p is the input parameter vector, and $\hat{S}$ is the predicted spectral output. Note that the hat notation denotes the predicted output by the model. The forward model learns to predict the THz spectrum $\hat{S}$ for a given parameter vector, with the mean squared error (MSE) loss function

$$Loss_{forward} = mean\left(w \times (\hat{S} - S)^2\right). \tag{2}$$

Here, w denotes a spectral weighting parameter utilized to attenuate geometry-induced Fabry–Pérot interference that randomly appears in the THz spectrum, thereby directing model learning toward spectrally relevant features by assigning a lower weight to the spectral region affected by unwanted Fabry–Pérot interference. ForwardNet is trained in advance by minimizing the spectral loss between the predicted and simulated transmittance spectra, and its parameters are fixed after convergence,

$$\hat{p} = \mathrm{InverseNet}(S), \tag{3}$$

where $\hat{p}$ is the predicted structural parameter vector for the target THz transmittance spectrum, S. The inverse model is trained under the tandem architecture using a pretrained and fixed ForwardNet. During the training procedure by the given target spectrum S, the InverseNet predicts a corresponding parameter vector $\hat{p}$, which is then passed through the fixed ForwardNet to reconstruct the $\hat{S}$. The inverse model was trained using a combined loss function that incorporates both the spectrum reconstruction loss and the parameter prediction loss. Here, the spectral data are bounded between 0 and 1, whereas the structural parameters are standardized using z-score normalization, resulting in different numerical scales. This discrepancy can cause the model to be biased toward minimizing the parameter loss, which typically has a higher numerical range. To ensure balanced contributions from both losses, we introduced a scaling factor as follows:

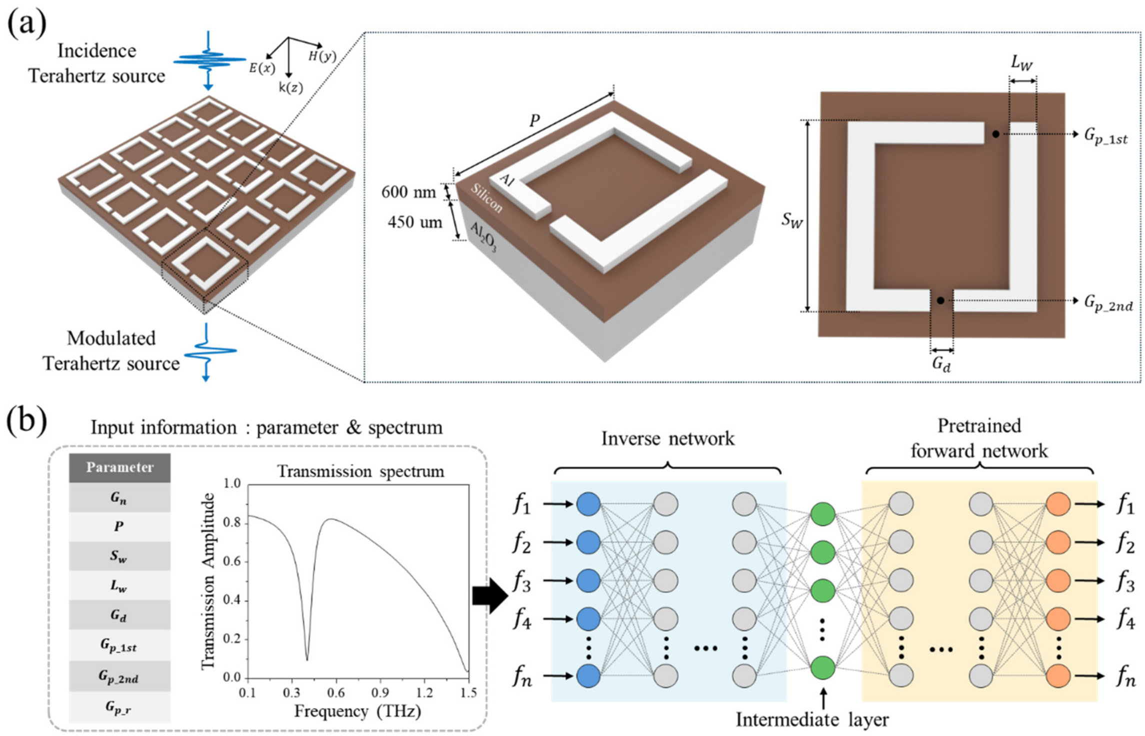


FIG. 1. Tandem network for THz modulator inverse design. (a) Geometry of split-ring resonator array and its corresponding cross sectional structure design with variable parameters. (b) Simplified representation of a tandem network with training data, which consists of paired split-ring resonator parameters and spectrum. A tandem network is composed of an inverse design network connected to a pretrained forward network. In the training process of the inverse network, weights in the pretrained forward network are fixed, and the weights in the inverse network are optimized.

$$\alpha_{epoch} = \frac{l_{param}}{l_{spec}}, \tag{4}$$

where l_{spec} is the MSE between the input target spectrum and the predicted spectrum by the fixed ForwardNet, and l_{param} is the MSE between the ground-truth and predicted parameters. The scaling factor is computed at each epoch to normalize the magnitude of the parameter loss to match that of the spectrum loss. During training, the combined loss is computed and then backpropagated to update the model parameters based on the gradients of the loss function, which determine the direction and magnitude of the updates to each trainable weight in the network. The gradient of the inverse loss is given as

$$\nabla Loss_{inverse} = (1-\lambda) \times \nabla l_{spec} + \lambda \times \frac{1}{\alpha_{epoch}} \times \nabla l_{param}, \tag{5}$$

where λ is the weighting ratio that adjusts the relative importance between the two loss terms. Initially, a value of $\lambda = 0.5$ was chosen to equally weight the spectrum and parameter losses during this learning process. Importantly, the gradients of l_{spec} and l_{param} are preserved independently, guiding the model in its respective direction. The scaling factor α_{epoch} only adjusts the magnitude of the losses without altering their gradient directions. This ensures that both losses contribute independently to the model updates, allowing the model to improve both aspects simultaneously without mutual interference, while maintaining a balanced contribution during training. All deep learning models were implemented using the PyTorch framework, which provides flexibility for neural architecture design and efficient GPU acceleration. Training was conducted using the Adam optimizer with an adaptive learning rate scheduler to enable efficient and stable convergence. The adaptive scheduler dynamically adjusts the learning rate during training, allowing faster progress during early stages and finer updates as the model approaches optimality (details on model optimization are provided in the supplementary material, Sec. 2). To assess model performance, the 13 000 spectrum-geometry samples were split into training (90%), validation (5%), and test (5%) sets. On the test set, the ForwardNet achieved an MSE of 0.0014, and when the predicted geometry parameters from the InverseNet were fed into the ForwardNet to generate reconstructed spectra, the resulting MSE was 0.0008. These results confirm that both models generalize well across the parameter space and accurately reproduce the target spectral responses.

The trained InverseNet was evaluated using arbitrarily generated target spectra. Each target spectrum was projected through the autoencoder to guide the network toward physically realizable solutions and then passed into the inverse model to retrieve the corresponding structural parameters (see the supplementary material, Sec. 3). These predicted parameters were used to construct SRR geometries, which were subsequently validated by CST Microwave Studio and compared with the autoencoded target spectrum, shown as the blue solid line in the THz spectra of Fig. 2. The simulated spectra based on the predicted parameters (orange dotted line) exhibited excellent agreement with the autoencoded input target spectra across all cases, demonstrating the high fidelity and effectiveness of the inverse design process. Additionally, the predicted spectra were quantitatively evaluated against the target spectra across 45 representative test cases, covering variations in single-peak and dual-peak resonances, using cosine similarity, correlation coefficient, and MSE. Across representative test cases, the predictions achieved a cosine similarity around 0.98, correlation coefficients between 0.86 and 0.96, and MSE values mostly below 0.01, indicating that the model accurately captures both the center frequency and linewidth variations of the target spectra (see the supplementary material, Sec. 5). In addition, we investigated that InverseNet can capture the inherent multi-solution of the inverse design problem. The inverse design problem is fundamentally ill-posed because multiple distinct structural configurations can produce nearly identical spectral responses. This non-uniqueness referred to as multi-solutions poses both a challenge and an opportunity in photonic design. While it complicates direct parameter retrieval, it also broadens the feasible design space and enables greater flexibility in practical applications. Therefore, assessing whether an inverse model can identify physically consistent yet structurally diverse solutions beyond the training distribution is crucial for validating its robustness and generalizability. To verify the existence of multi-solutions, we evaluated the model using target spectra generated from geometry parameters that were deliberately excluded from the training dataset. Two representative test cases were selected, in which input spectra were synthesized from known ground-truth geometry parameters (Fig. 3).

When these spectra were passed through the trained inverse model, the predicted parameters deviated notably from the ground-truth geometry parameter. However, the simulated spectra of the predicted designs closely matched the target responses. This demonstrates strong spectral consistency despite structural differences, confirming the presence of multiple valid geometries that result in similar optical outputs. This result provides compelling evidence that multi-solutions exist in inverse design, where distinct geometries can yield similar spectral responses. Mathematically, this reflects a many-to-one mapping in spectral space, such that $\mathrm{ForwardNet}(\hat{p}_1) \approx \mathrm{ForwardNet}(\hat{p}_2)$ despite $\hat{p}_1 \neq \hat{p}_2$. Importantly, the ability of the inverse model to retrieve distinct but forward consistent designs indicates that it has learned a generalized mapping from spectrum to structure, rather than memorizing specific training instances. This generalization is essential in practical photonic design scenarios, where a single target spectrum may correspond to a manifold of valid configurations. Thus, the results illustrated in Fig. 3 not only validate the physical plausibility of the inverse predictions but also demonstrate the model's ability to capture the underlying spectral-structural relationship, even in the presence of inherent non-uniqueness.

High-Q factors, sharp resonance, and strong field enhancement are essential design requirements for achieving high depth modulation in THz amplitude modulators. To demonstrate that our inverse design framework satisfies these demands not only by reproducing target spectra but also by generating physically meaningful geometries based

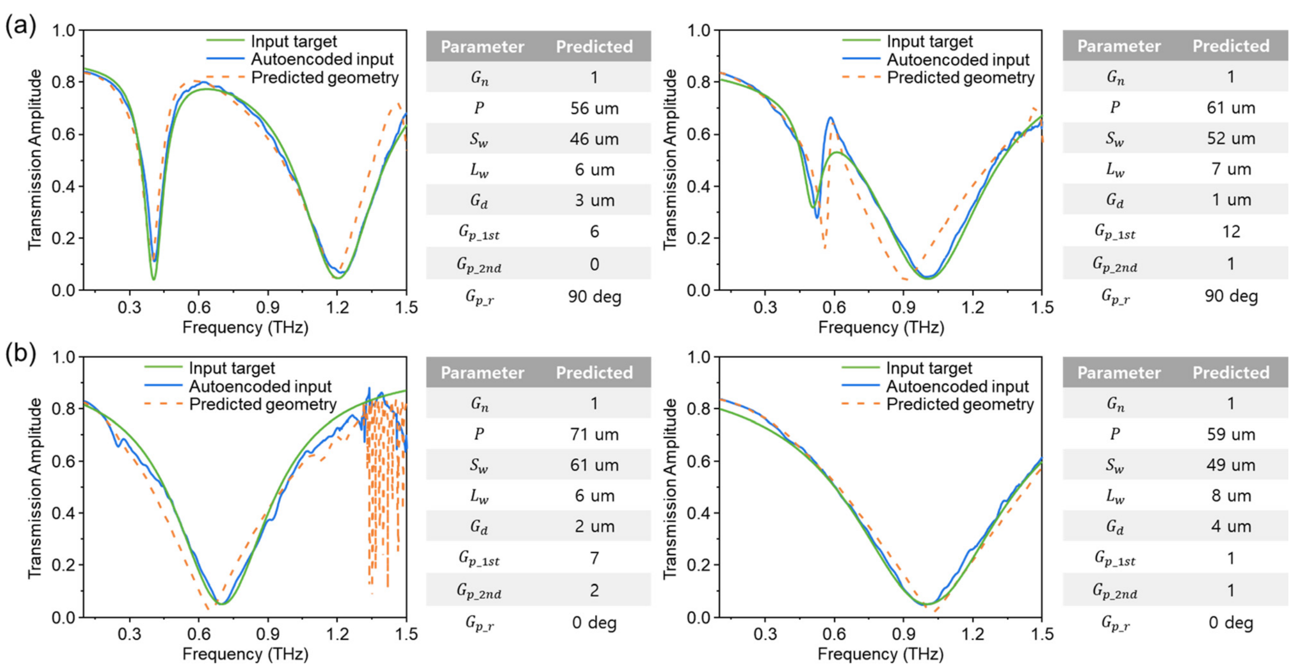
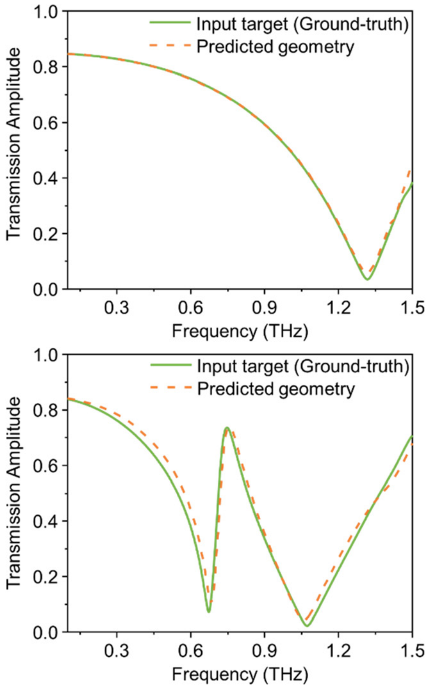


FIG. 2. Inverse design performance of InverseNet. Each panel presents an input target spectrum (green), autoencoded input target spectrum (blue), the structural parameters of the predicted SRR geometry (shown in the table on the right), and the simulated spectrum (orange dots) based on the predicted parameters of InverseNet. The simulated spectra show good agreement with the target spectra, exhibiting a single peak (a) and dual peaks (b). An input target spectral profile was synthesized by applying a Lorentzian oscillator model, with resonance parameters as listed in the supplementary material, Table S10.



Parameter	Ground-truth	Predicted
G_n	2	2
P	55 um	60 um
S_w	45 um	49 um
L_w	6 um	7 um
G_d	2 um	1 um
G_{p_1st}	9	6
G_{p_2nd}	13	-4
G_{p_r}	90 deg	90 deg

Parameter	Ground-truth	Predicted
G_n	2	2
P	60 um	61 um
S_w	50 um	49 um
L_w	8 um	7 um
G_d	6 um	6 um
G_{p_1st}	6	-12
G_{p_2nd}	10	-3
G_{p_r}	0 deg	0 deg

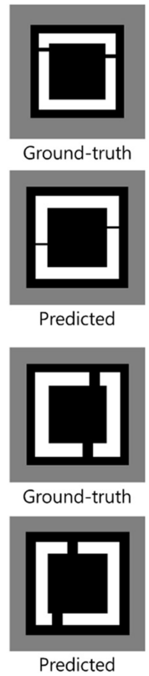


FIG. 3. Validation of multi-solutions in the InverseNet. Each panel shows the ground-truth THz spectrum (green), the EM simulated spectrum generated from the inverse design geometries (orange dots), and a comparison of the ground-truth and predicted geometrical parameters (right panels). The inverse model receives the ground-truth spectrum as an input and generates predicted geometrical sets, which are then EM simulated to verify the presence of a multi-solution mapping between different geometries producing nearly identical spectra.

on distinct resonance mechanisms, we applied it to a previously reported Fano resonant modulation spectrum as a representative benchmark case.[17] As shown in Fig. 4(a), the asymmetric and sharp dip of the Fano spectrum (green solid line) was first projected into the latent space via a trained autoencoder, yielding an autoencoded spectrum (blue solid line) that retains the key resonance features. This latent representation was then passed into the inverse model, which predicted the structural parameters expected to reproduce the input target spectrum. Interestingly, the predicted structure not only differs from the previously reported design but also lies outside the parameter range covered by the training data. Nevertheless, the simulated modulation spectrum (orange dotted line) derived from this structure still preserves the hallmark features of Fano resonance, including a sharp and asymmetric resonance dip. This result highlights the model's ability to effectively learn the underlying relationship between spectral features and structural configurations. To further assess the modulation performance of the predicted Fano resonant structure, we conducted simulations assuming photoexcitation via optical pumping, in which the electrical conductivity (σ_{Si}) of the silicon layer was systematically varied. Photoexcitation of the 600 nm Si layer was modeled by assigning different constant electrical conductivities in CST material setting. In this formulation, the complex permittivity is expressed as

$$\varepsilon(\omega) = \varepsilon_{Si} - j\frac{\sigma_{Si}}{\omega\varepsilon_0}, \tag{6}$$

where ε_{Si} is the permittivity of silicon, σ_{Si} is the electrical conductivity of silicon, and ε_0 is the vacuum permittivity. The conductivity was swept over a range consistent with photocarrier generation by assuming optical pumping. This steady-state conductivity-based approach is widely used for THz active metamaterial simulations, as the resonance tuning is dominated by carrier-induced conductivity changes rather than transient carrier dynamics. As shown in Fig. 4(b), increasing σ_{Si} resulted in a gradual reduction in resonance depth, effectively reproducing the modulation behavior driven by photoconductivity. These results confirm that the inverse-designed geometry possesses the essential dynamic response characteristics required for active THz modulation under optical excitation. Figure 4(c) presents the electric field distribution in the predicted Fano resonant SRR geometry. At the Fano resonance frequency, a strong localization of the electric field is observed at the split gap, which arises from capacitive effects inherent to the SRR structure. The accumulation of oscillating charges across the gap enhances the local field intensity, effectively forming a capacitive hotspot. Under photoexcitation, the increased conductivity of the silicon layer suppresses this capacitive effect, resulting in the disappearance of the localized electric field. The corresponding surface current distributions shown in Fig. 4(d) further supports this behavior. In the unpumped state, the SRR exhibits anti-parallel surface current flow at resonance, indicative of a magnetic dipole mode oriented normal to the plane of the structure. This anti-parallel current pattern represents a non-radiative dark mode, and its interference with the bright LC-type dipole mode supported by the SRR forms the asymmetric Fano resonance observed in the transmission spectrum. Upon photoexcitation, this current distribution transitions from anti-parallel to parallel, disrupting the interference condition necessary for Fano resonance and consequently diminishing the sharp resonance dip. Additional electric field and surface current distributions at various frequencies are presented in Fig. S9 in the supplementary material, further elucidating the resonance dynamics and supporting the observed modulation behavior (see the supplementary material, Sec. 4).

In this study, we presented a tandem network-based inverse design framework for SRR-based THz amplitude modulators. The

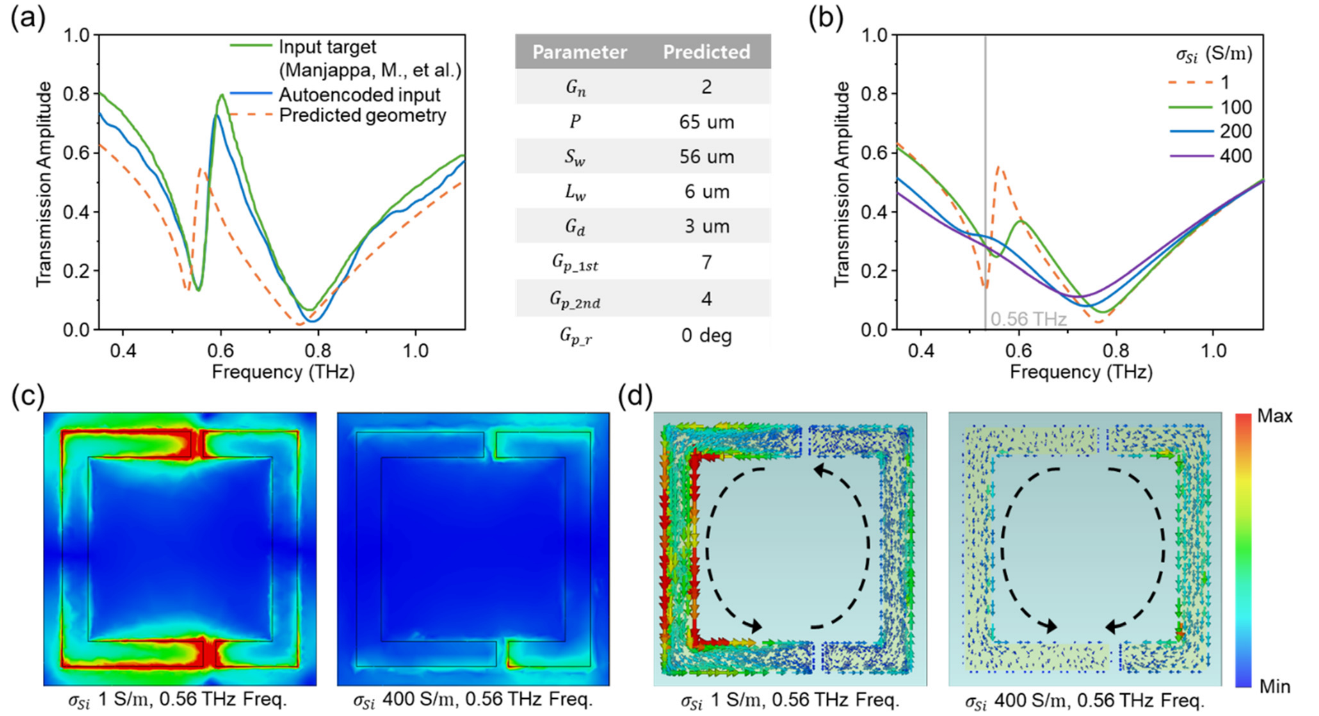


FIG. 4. Inverse design and field analysis of Fano resonant SRR geometry. (a) Inverse design based on a previously reported Fano resonance spectrum (green), along with its latent space projection via an autoencoder (blue).[17] The structural parameters predicted by InverseNet are listed in the accompanying table, and the corresponding simulated spectrum (orange dots). (b) Simulated amplitude transmission spectra as a function of silicon conductivity (σ_{Si}), modeling the effect of optical pumping. (c) Electric field distributions at the Fano resonance dip, shown for both the unpumped and photoexcited states. Upon photoexcitation, enhanced conductivity effectively suppresses the field in the SRR gap. (d) Surface current distributions at the Fano resonance dip. The photoinduced switching results in a transition from antiparallel to parallel current configurations, confirming the modulation effect at the structural level. Reproduced with permission from Manjappa *et al.*, Adv. Mater. **29**, 1603355 (2017). Copyright 2017 Wiley-VCH GmbH.

framework integrates a pretrained forward model, an inverse network, and an autoencoder, enabling consistent and physically plausible inverse predictions. Although tandem architectures have been widely used in previous optical inverse design studies, their combination with an autoencoder projection module provides distinct advantages in the THz domain, where multi-solution mappings and distribution mismatch are particularly severe. This hybrid framework ensures physically consistent inverse predictions and significantly stabilizes the retrieval of high-Q THz resonances that are challenging to model using direct or implicit inverse networks. Comprehensive evaluations across diverse synthetic target spectra confirm that the model accurately reconstructs spectral features and generalizes beyond the training distribution (see the supplementary material, Sec. 5). Furthermore, to validate the physical interpretability and practical applicability of the model, we applied the framework to the inverse design of Fano resonant geometries with sharp, asymmetric spectral dips and high-Q resonances. The resulting structures not only reproduced the spectral features of Fano resonances but also exhibited consistent electric field localization and surface current distributions corresponding to magnetic dipole modes, as confirmed through full-wave simulations. These results demonstrate the model's capability to synthesize complex electromagnetic interactions and validate its potential for designing advanced THz metastructures. The full implementation, including code and datasets, is publicly available on GitHub to promote reproducibility and accelerate AI-driven inverse design in the THz domain, serving as a valuable resource that enables broader exploration of intelligent and reconfigurable THz photonic systems and supports future advances in AI-driven THz device design.

See the supplementary material for the following: Sec. 1 provides details on the training data preparation for deep learning, Sec. 2 provides details on the optimization of the tandem network, Sec. 3 provides details on autoencoder-based projection of the target spectrum, Sec. 4 provides inverse design and field analysis of Fano resonance geometries using Lorentz-oscillator models, and Sec. 5 provides statistical evaluation of inverse design performance.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Tae-In Jeong and Eunji Choi contributed equally to this paper.

Tae-In Jeong: Conceptualization (equal); Methodology (lead); Software (equal); Writing – original draft (lead). **Eunji Choi:** Formal analysis (lead); Methodology (lead); Software (lead); Writing – original draft (lead). **Robert A. Taylor:** Funding acquisition (equal); Project administration (equal); Writing – review & editing (equal). **Seungchul Kim:** Conceptualization (equal); Funding acquisition (lead); Project administration (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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